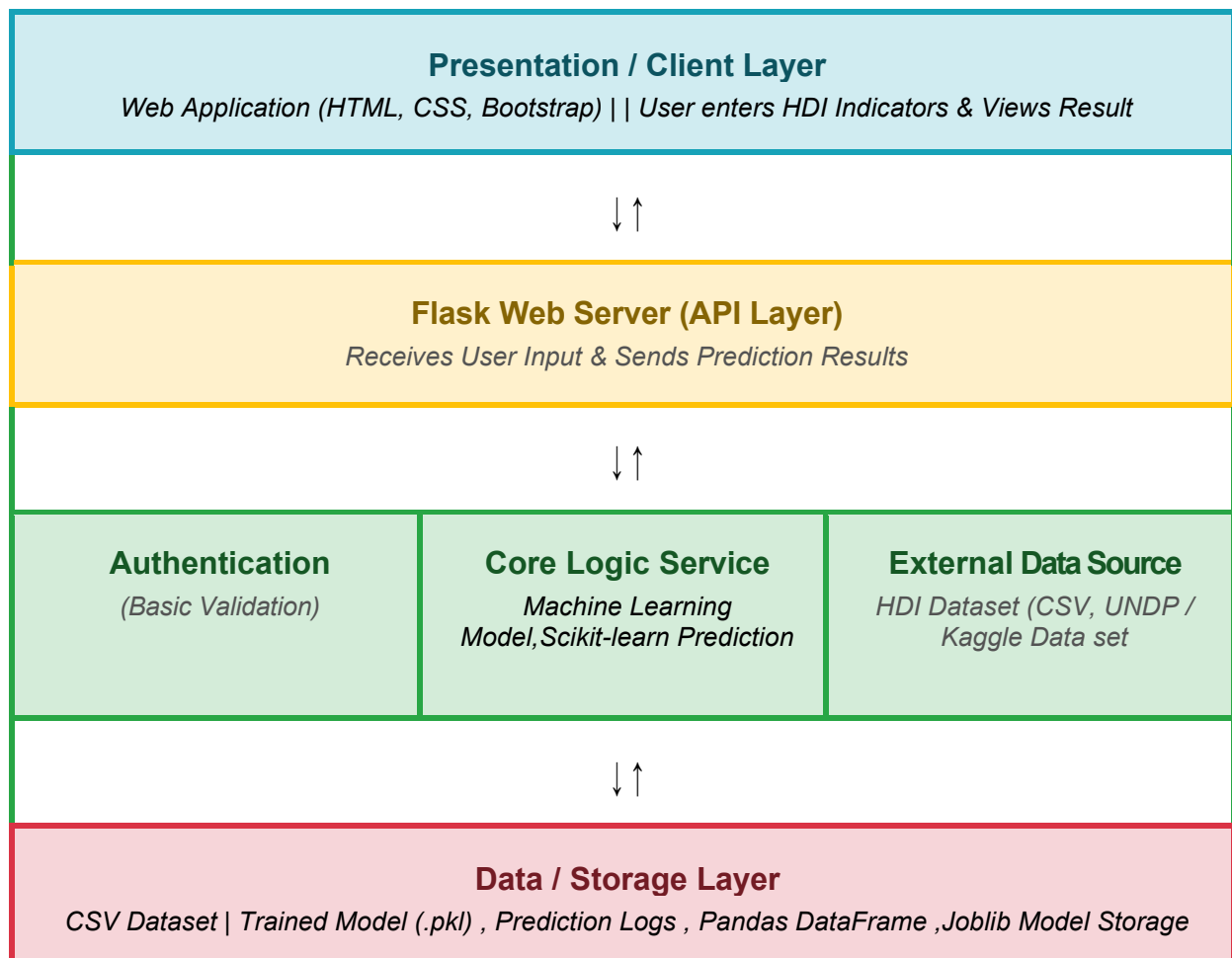


Solution Architecture

Date	15 March 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides a user-friendly web interface where users enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita to predict the HDI category.</i>	<i>HTML5, CSS3, Bootstrap</i>
<i>API Gateway</i>	<i>Receives user requests, validates the input, forwards the data to the machine learning model, and returns the predicted HDI category to the user.</i>	<i>Flask (Python Web Framework)</i>
<i>Core Logic Service (Machine Learning)</i>	<i>Processes the input data using the trained machine learning model and predicts the Human Development Index (HDI) category (Very High, High, Medium, or Low).</i>	<i>Python, Scikit-learn, Pandas, NumPy, Joblib</i>
<i>External Data Source</i>	<i>Provides the HDI dataset used for training and evaluating the machine learning model.</i>	<i>UNDP Dataset / Kaggle Dataset (CSV)</i>
<i>Database / Data Layer</i>	<i>Stores the HDI dataset, trained machine learning model, and prediction-related information required for the application.</i>	<i>CSV File, Pandas DataFrame, Joblib (.pkl)</i>